Mathematics 2, part 4, lecture 3

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- Linear programming
  - ~~LP toy problem~~
  - ~~Problems with LP problems~~
  - ~~LP duality~~
  - ~~Dimension analysis~~
  - ~~LP interior point methods, why?~~
  - GD in interior point methods
  - When to stop descending?
  - How to start?
  - Time complexity.

Invariants in the iteration

In every step we have

(I1) (primal strict feasibility) $Ax = b$ with $x > 0$

(I2) (dual strict feasibility) $A^T y + s = c$ with $s > 0$

(I3) $\sigma^2 := \sum_i \left(\frac{x_i s_i}{\mu} - 1 \right)^2 \leq \frac{1}{36}$.

We can take $\mu' = \left(1 - \frac{1}{6\sqrt{m}}\right)\mu$.

Final rounding

Assumption: Let L be a positive number so that every nonzero coordinate in an optimal solution of (P) or (D) is at least L.

Lemma

Let (x, y, s) *and* μ *satisfy the invariants (I1), (I2), and (I3). Le* x^* *be an optimal solution to (P) and let* (y^*, s^*) *be an optimal solution to (D). Assume that the smallest nonzero value of* x^* *and* s^* *is at least L.*

- *If* $x_i < L/(4m)$ *then* $x_i^* = 0$ *in every optimal solution and*
- *if* $s_i < L/(4m)$ *then* $s_i^* = 0$ *in every optimal solution.*

Theorem (strong duality)

For every i *we have* $x_i^* = 0$ *in every optimal solution or* $s_i^* = 0$ *in every optimal solution. Thus*
$c^T x^* - b^T y^* = (x^*)^T s^* = 0.$

Initial solution (traditional)

$$\begin{aligned} \min\ & 480x_1 + 600x_2 \\ & 2x_1 + 3x_2 \geq 3 \\ & 2x_1 + 1x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} \min\ & 480x_1 + 600x_2 \\ & 2x_1 + 3x_2 \geq 3 - x_0 \\ & 2x_1 + 1x_2 \geq 2 - x_0 \\ & x_1, x_2, x_0 \geq 0 \end{aligned}$$

$$\begin{aligned} \min\ & x_0 \\ & 2x_1 + 3x_2 \geq 3 - x_0 \\ & 2x_1 + 1x_2 \geq 2 - x_0 \\ & x_1, x_2, x_0 \geq 0 \end{aligned}$$

Initial solution

We want to transform (P) — by possibly adding additional variables — to the *artificial problem* (art P) so that

- the feasible domain is bounded (preserving the optimum if it exists),
- flagging the possible infeasible problem,
- so that it is possible to find strictly feasible solutions to (art P) and its dual (D),
- which also satisfy the invariant (I3) for a suitable initial choice of μ_{init}.

Complexity of IP

Theorem

The interior point method (as described during lectures) terminates in polynomial number of steps.

LP, historical notes

- mid 19th century, Fourier: elimination of variables from a system of linear inequalities.
- 1940s, Dantzig: simplex algorithm (worst case exponential)
- 1940s, von Neuman: duality
- 1970s: ellipsoid method, first polynomial algorithm (not practical)
- 1984, Karmarkar: first interior-point algorithm (polynomial, practical)
- still open: is there a strongly polynomial algorithm form LP.

Bottom line: interior point methods work well for millions of variables+constraints, often with a fixed number (less than 100) iterations.

Interior point methods not just for linear programming

- quadratic programming

$$\begin{aligned} \min \; & x^T Q^T Q x + c^T x \\ & Ax \leq b \end{aligned}$$

- semidefinite programming

$$\begin{aligned} \min \; & \mathrm{trace}(C^T X) \\ & \mathrm{trace}(A_i^T X) \leq b_i \\ & X \text{ positive semidefinite} \end{aligned}$$

Interior point methods not just for continuous optimization

Let G be a graph. A *matching M* is a set of edges without common endvertices.

MaximalMatching

$$\max \sum_e x_e$$
$$\text{for all } v : \sum_{v \sim e} x_e \leq 1$$
$$x \in \{0,1\}^{E(G)}$$

lin-relax MaximalMatching

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$$\text{for all } v : \sum_{v \sim e} x_e \leq 1$$
$$x \in [0,1]^{E(G)}$$